

SNAPDEAL CUSTOMER INSIGHTS REPORT

(Market Basket Analysis with Python)

PROJECT SUMMARY

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EXECUTIVE SUMMARY

This report summarizes the analysis of Snapdeal's customer purchasing behaviour survey data with 800 respondents and 24 fields. The goal was to clean the dataset, understand purchasing patterns, segment customers and get actionable insights on recommendations, reviews and retention.

DATASET NOTE- Source column descriptions contain 'Amazon' references from the original survey template. Though, analysis has been done throughout the project under the Snapdeal framing per the assignments brief.

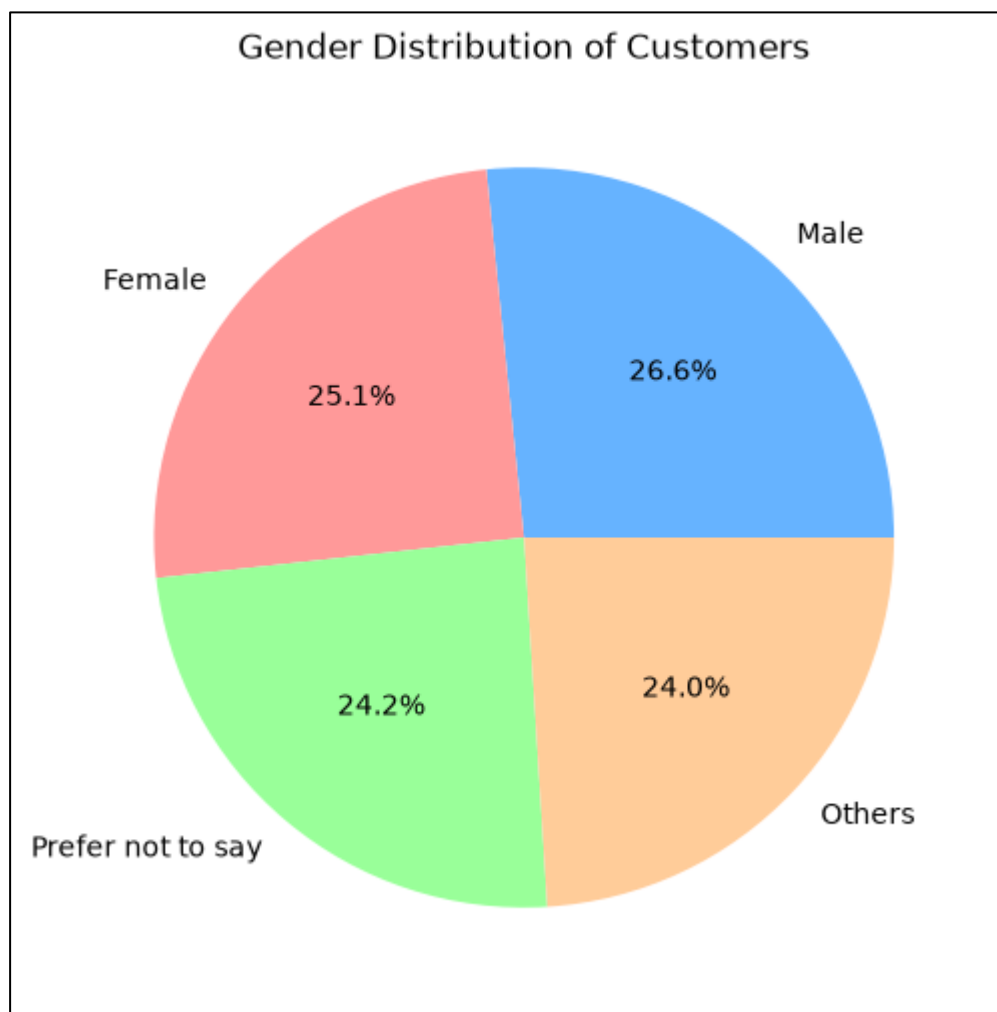
TASK 1- DATA CLEANING AND PREPARATION

Missing and junk data clustered specifically around open-ended and questioned fields (search method, service appreciation, improvement areas).

ISSUE FOUND	FIX APPLIED
Duplicate Column name- Personalized_Recommendation_Frequency appeared twice	Renamed numeric version to *_Numeric
0 duplicate rows	Verified through duplicated (). sum ().
Product_Search_Method-149 missing+Inconsistent casing	Standardized casing, filled missing as 'Unknown'
Hidden trailing whitespace across text columns	Regex word-boundary replace to 'Others'.
'.' Used as junk/placeholder answers (89+38 rows)	Replaced with NaN, then filled as 'Not Specified'
Numeric rating columns	Verified already correct int64 dtype

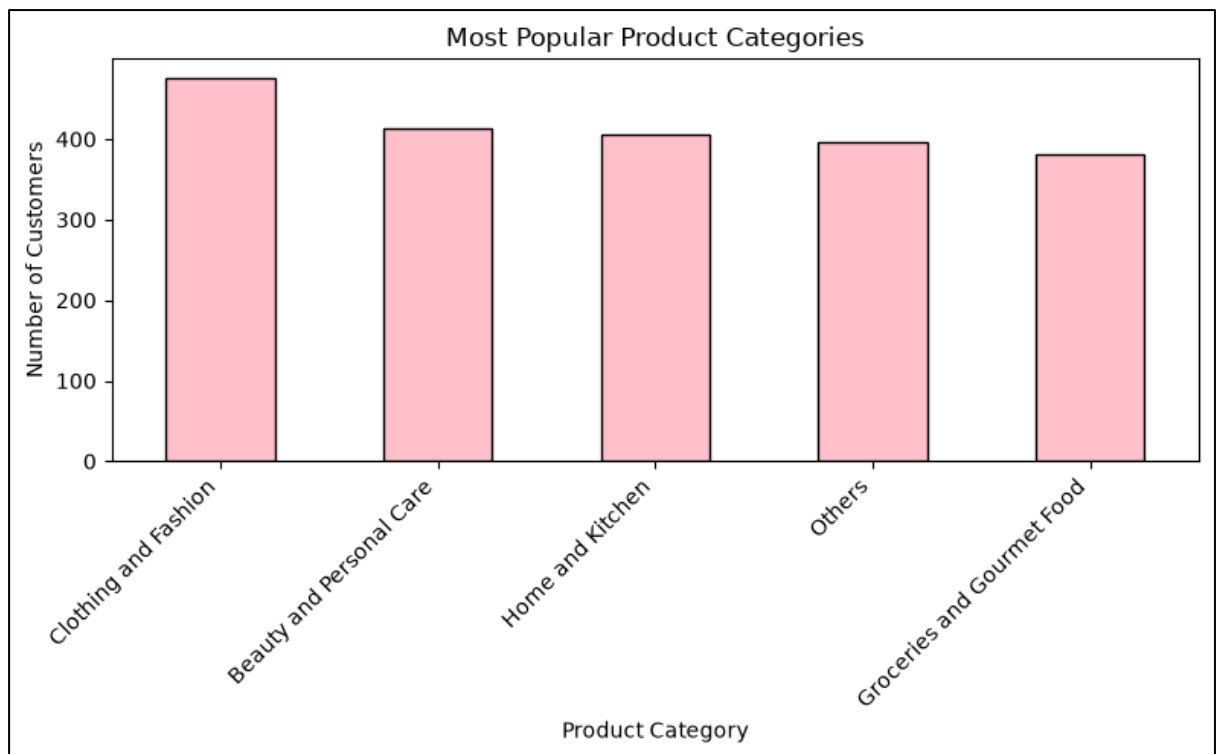
TASK 2- DESCRIPTIVE BEHAVIOR ANALYSIS DEMOGRAPHICS-

- AGE- Distribution is close to uniform across 3-67 with no natural clustering, where 114 respondents are under age 13, a strong signal this field was randomly generated rather than real-world dataset. Therefore, Age has been treated as an unreliable variable throughout the analysis.
- GENDER-No dominant group has been found.



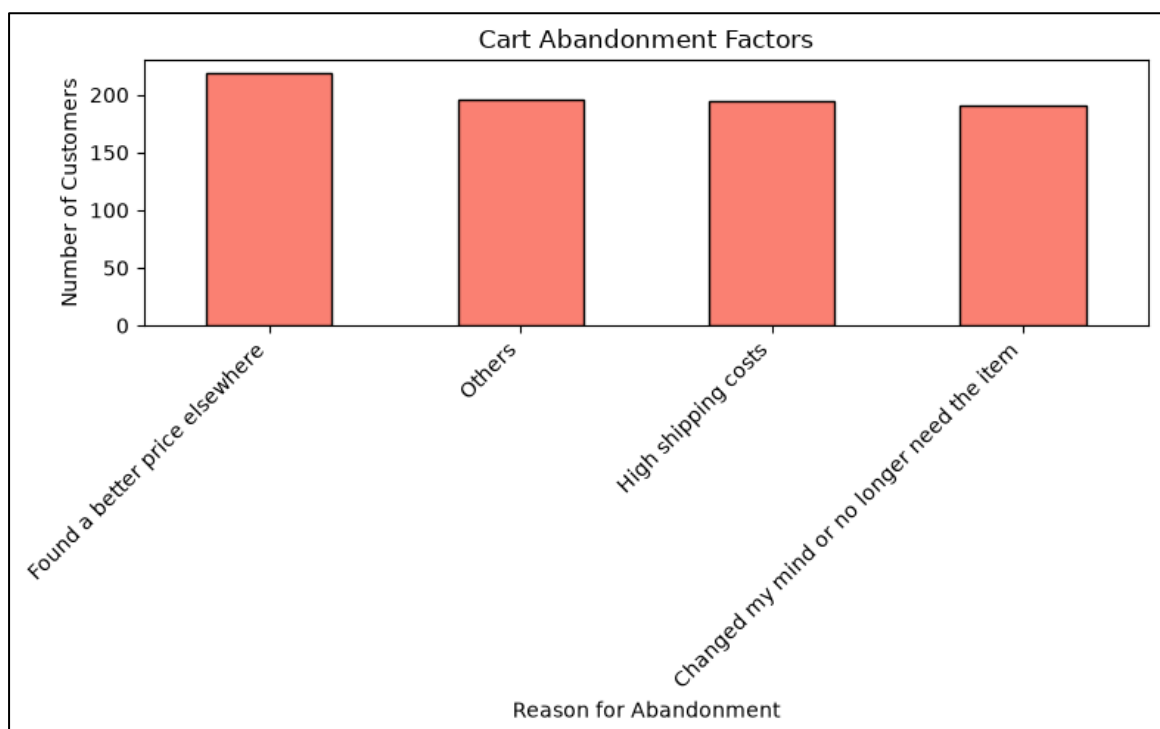
PURCHASE BEHAVIOR-

- Purchase frequency is evenly spread across all 5 categories.
- Clothing and Fashion with 476 mentions is the top product category followed by Beauty and Personal care with 414 mentions.



BROWSING AND CART BEHAVIOR-

The leading cause of cart abandonment is price competitiveness- 27.4% of customers abandon their cart after finding price elsewhere, ahead of high shipping cost and changed intent. This is one of the most actionable insights in Task 2.

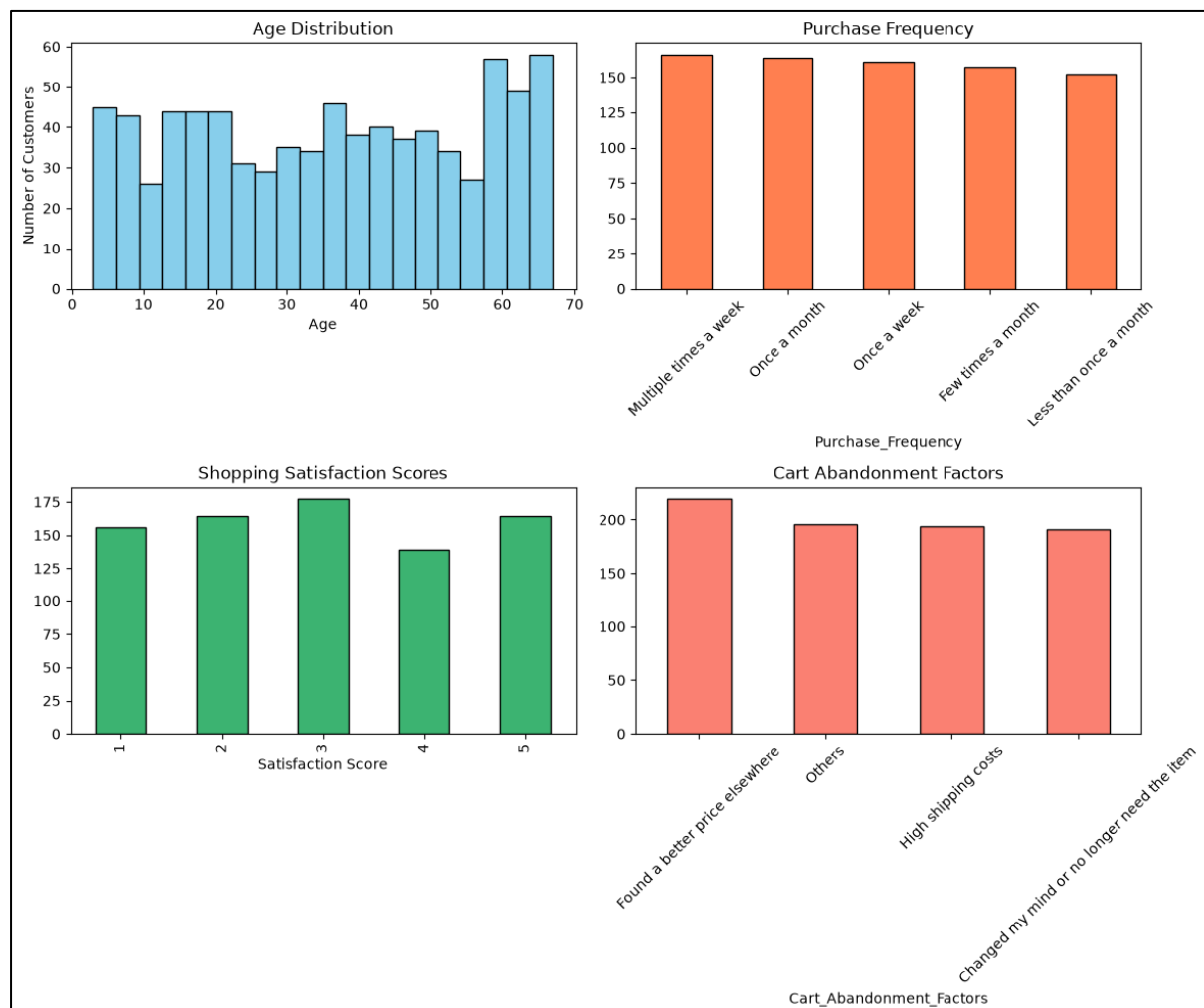


SATISFACTION METRICS-

All rating scale variables cluster around a neutral mid-point, which indicated customers are moderately satisfied, not strongly +ve or -ve.

Recommendation Helpfulness skews negative like 36.9% say recommendations are 'No' vs 32.6% 'Yes'.

METRIC	MEAN	MEDIAN
Shopping Satisfaction	2.99	3.0
Rating Accuracy	2.96	3.0
Customer Review Importance	2.99	3.0



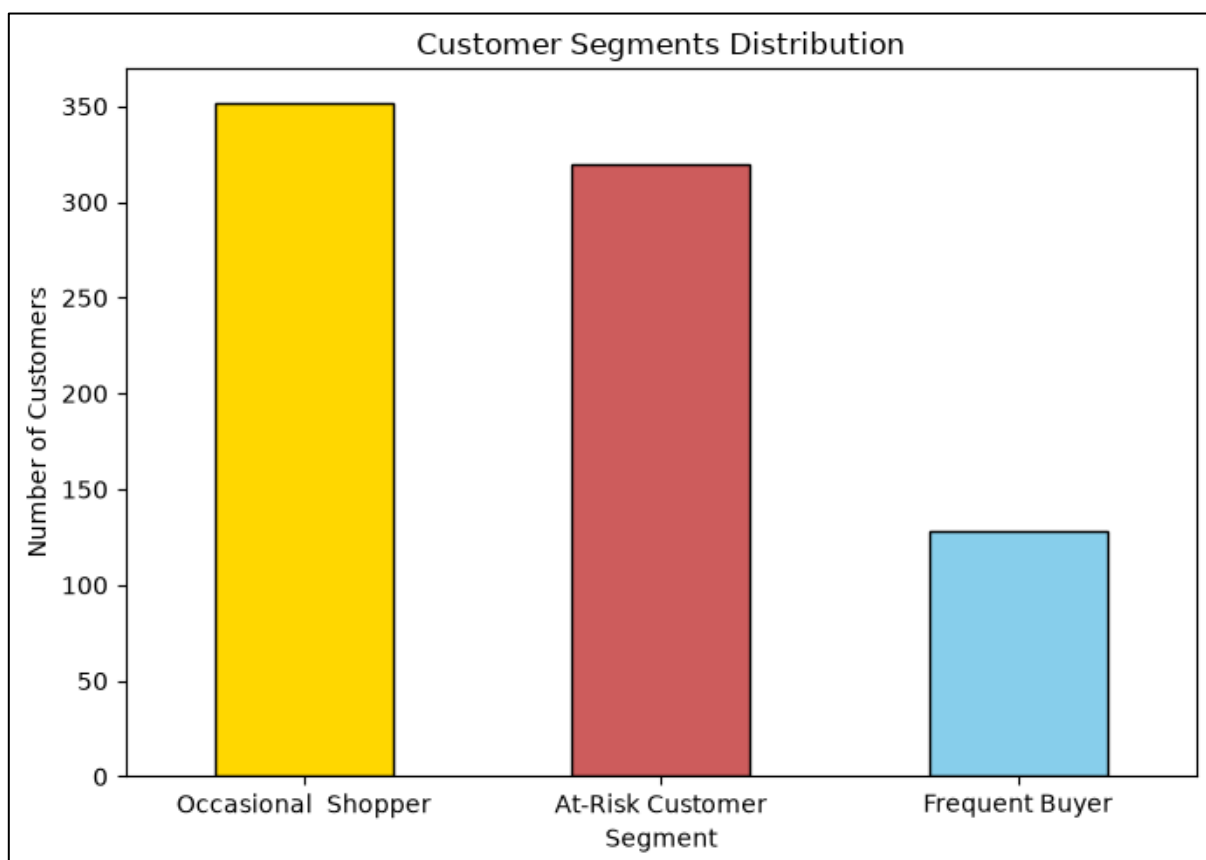
TASK 3- CUSTOMER SEGMENTATION AND PROFILING

Rules- Customers are segmented using purchase frequency (from 1-5 scale) and shopping satisfaction, with the following defined criteria

- At-Risk Customers- Shopping Satisfaction ≤ 2
- Frequent Buyer- Purchase Frequency ≥ 4 AND Shopping Satisfaction ≥ 4 .
- Occasional shopper- everyone else.

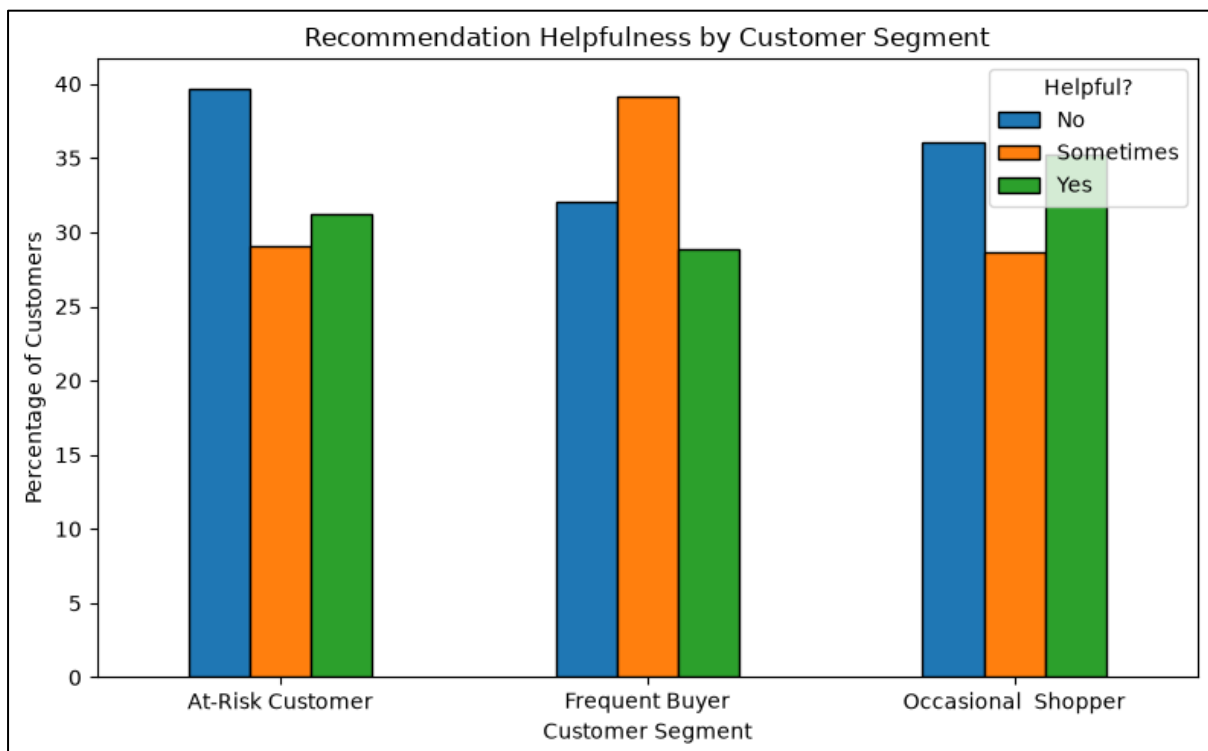
SEGMENT	COUNT	% OF CUSTOMERS
Occasional shopper	352	44.0
At-Risk Customers	320	40.0
Frequent Buyer	128	16.0

It's clear that more customers are At-Risk than are Frequent Buyers. This imbalance indicates Snapdeal's retention problem outweighs its current loyalty base.



DEMOGRAPHIC AND BEHAVIORAL DIFFERENCES ACROSS SEGMENTS-

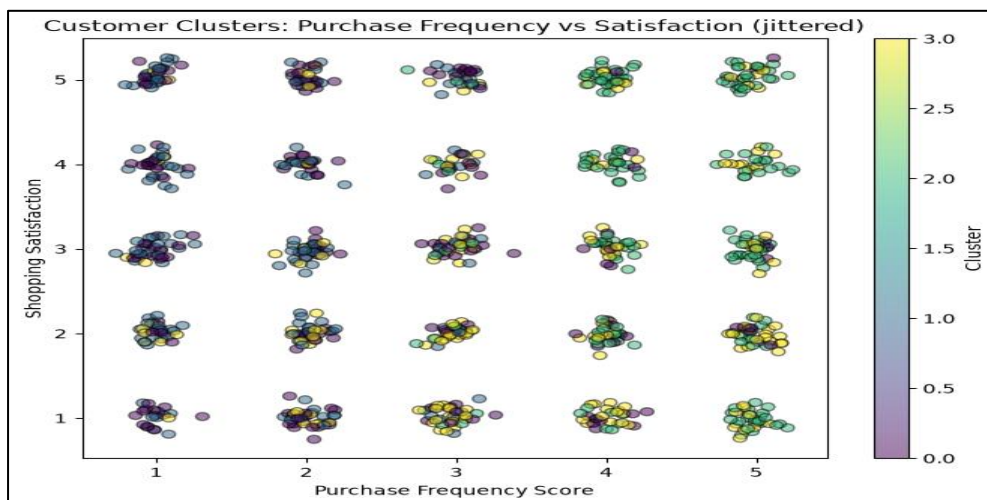
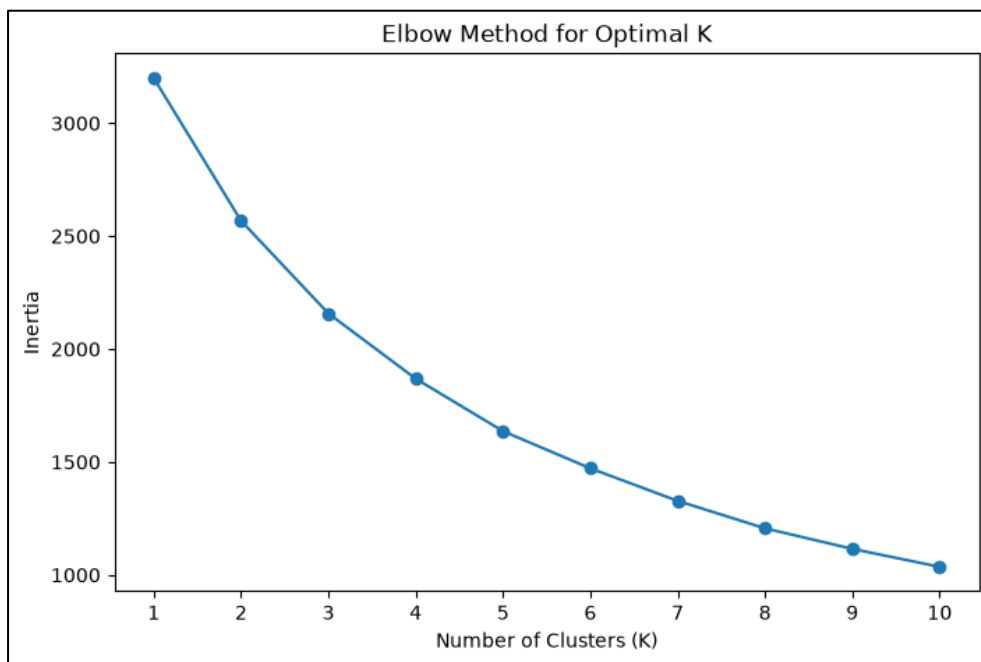
- Gender- This shows consistent proportions across all three segments. So, churn risk is not affected by gender.
- Age- This is virtually identical across segments (36-36.2). Also, this reinforces that age carries no real signal in this dataset.
- Cart abandonment reasons- It is also similarly distributed across segments, this is a platform-wide issue and not specific to At-Risk customers, few fixes will benefit all segments roughly equally.



Recommendation helpfulness is the one variable that differs by segment. At-Risk customers show the highest 'No' rate vs Frequent Buyers. This links poor recommendation quality to churn risk, directly.

K-MEANS CLUSTERING(UNSUPERVISED)-

- For the rule-based segments, K-means clustering has been applied to all the behavioural features- Purchase_Frequency_Score, Shopping Satisfaction, Customer Reviews Importance and Rating Accuracy. Henceforth, the optimal number of clusters (K=4) was selected using the Elbow Method and then identified the point where additional clusters yield diminishing improvement in fit.



INSIGHT- K-Means uncovered a pattern the manual rules missed that is frequency and satisfaction don't always go hand-in-hand together. At-Risk Customers buy fairly often yet report the lowest satisfaction than any group despite trusting ratings highly.

TASK 4- Recommendation and Review Insights-

- No trend was found between the frequency of personalized recommendations and how helpful customers find them (scores are narrowly ranged from 1.86–2.06 in a 1–3 scale), displaying recommendations more frequently does not make them feel more useful.

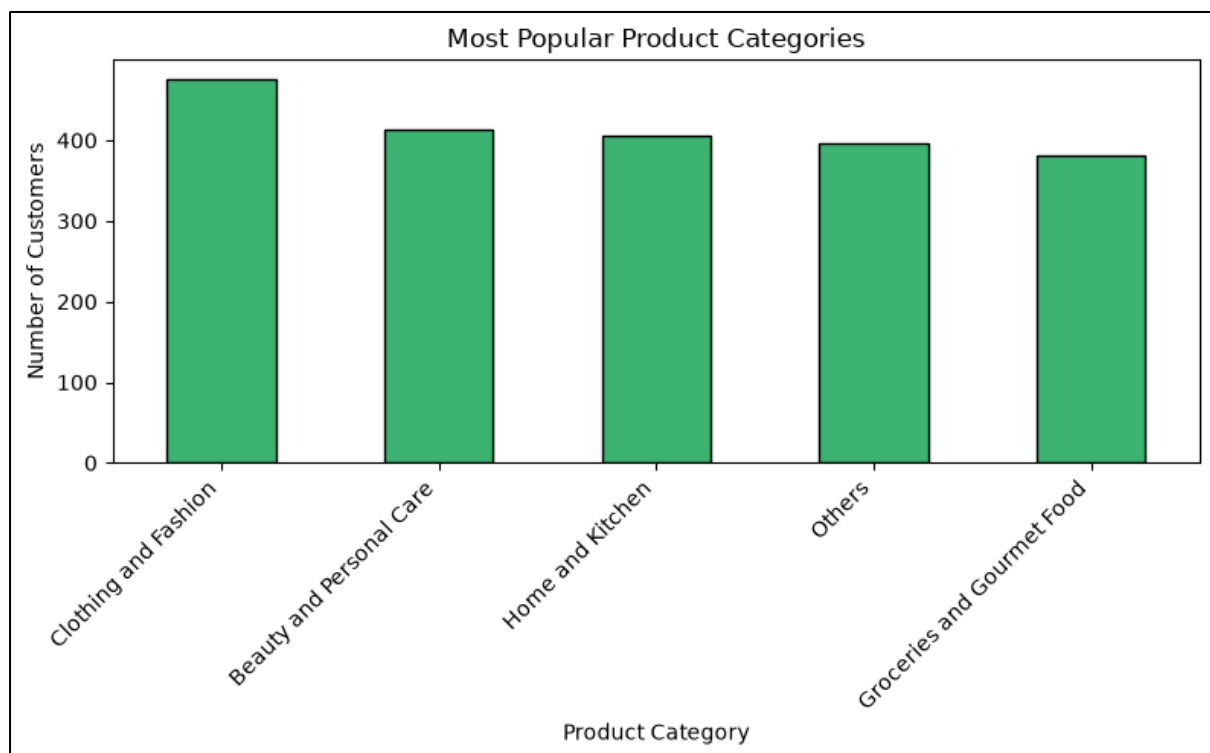
The three systems in this dataset, Insight Recommendations, reviews, and ratings are largely independent of each other, and trust in each has a different effect on the others. Improving one is unlikely to improve the others. The best fix to improve satisfaction with recommendations (not frequency) is to improve recommendation relevance (initially in the At-Risk segment), based on the fact that At-Risk customers are the least satisfied with recommendations specifically, and the findings of Task 3. Suggestions for the business.

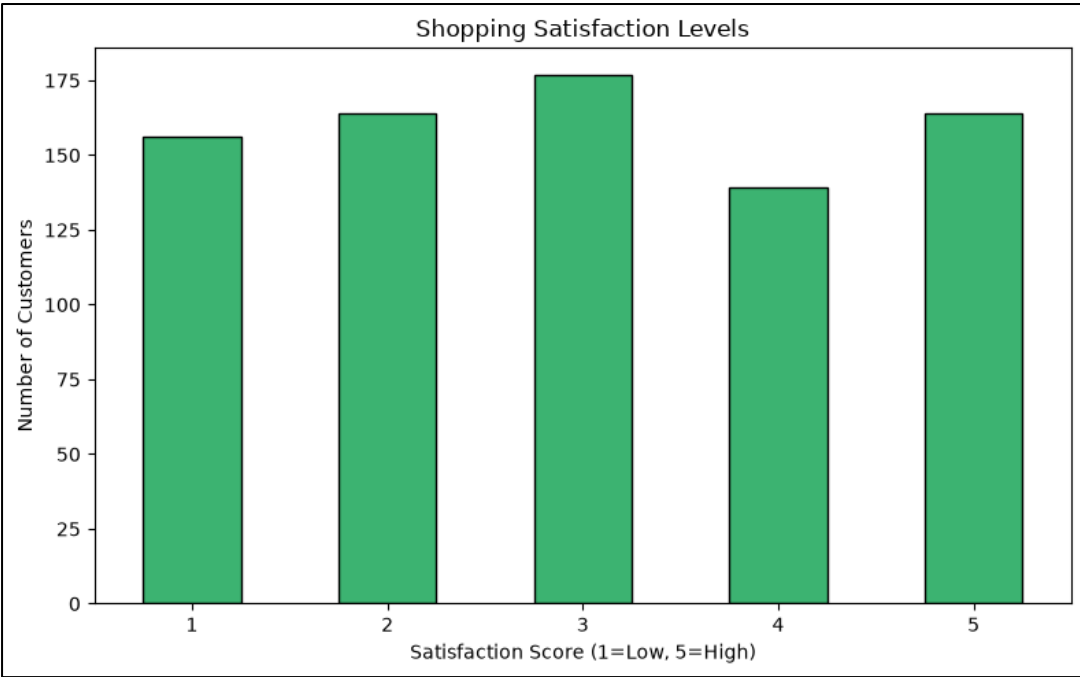
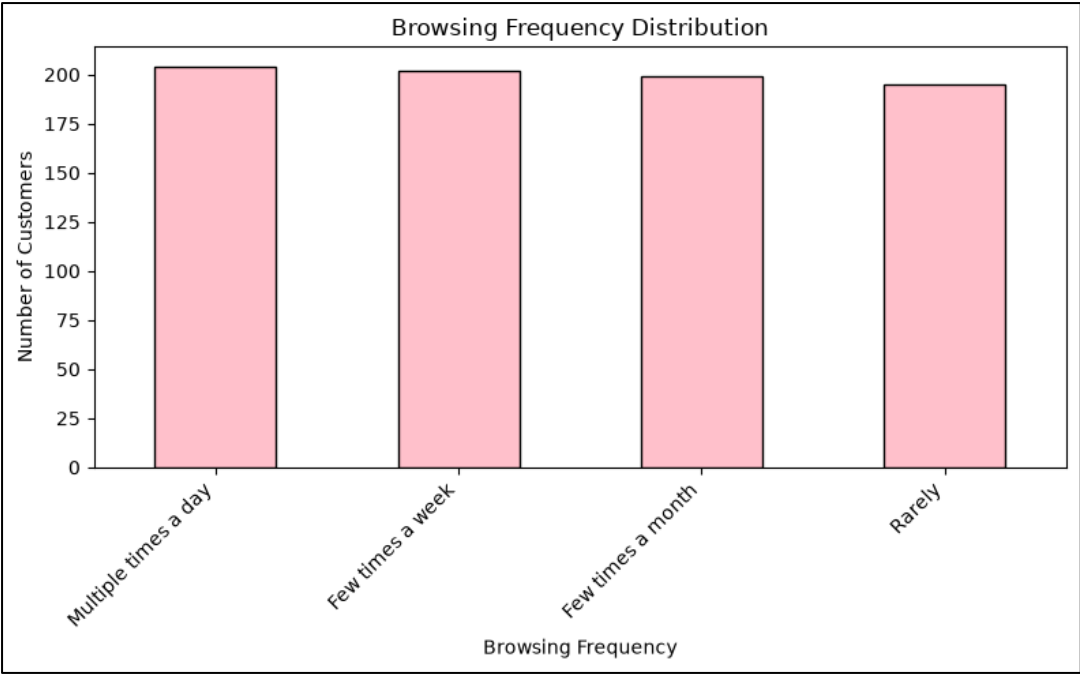
Recommendations for the Business-

- Recommendation quality and overall satisfaction are different problems – fixing the recommendation engine alone will not improve overall satisfaction.
- Relevance over frequency: showing recommendations more often has not shown to be beneficial; the algorithm needs to be better targeted, not more exposure.
- Importance of audit recommendation specifically for At-Risk / lower-satisfaction customers first as they have both the highest dissatisfaction with recommendations and highest churn risk.
- Treat review reliability, review helpfulness, and rating accuracy as separate levers with their own improvement efforts.

TASK 5- VISUALIZATION

Supporting visualizations were produced for: product category popularity, browsing frequency distribution, shopping satisfaction levels, and the recommendation-usefulness-vs-satisfaction relationship. Full code for all charts is included in the accompanying Jupyter notebooks (Task_2.ipynb, Task_3.ipynb, Task_5.ipynb).





OVERALL CONCLUSIONS & PRIORITIES

- Pricing competitiveness is the top lever for reducing cart abandonment, a platform-wide fix benefiting all segments equally.
- Customer retention (At-Risk segment, 40% of the base) is a bigger opportunity than loyalty growth (Frequent Buyers, 16%).
- Recommendation relevance and not frequency should be improved, prioritizing At-Risk customers first.
- Age data should not be used for targeting or segmentation decisions, as it appears to be randomly generated rather than reflective of real customers.
- K-Means clustering revealed a specific, high-value churn signal ('At-Risk Regulars') not captured by simple satisfaction-based rules, and is recommended as a starting point for any future targeted retention campaign.

VIDEO LINK- <https://drive.google.com/file/d/1m-vbQVtOQ95nIPPJU5lYqYc5pRxLTR9x/view?usp=sharing>